MIL-S-36040A
10 December 1975
SUPERSEDING
MIL-S-36040
13 February 1961

MILITARY SPECIFICATION

SPLINT, LEG, THOMAS, HALF-RING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a telescoping splint, suitable for use in immobilization of fractures and treatment of injuries to the lower extremities.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-225/8	-	Aluminum Alloy, Bar, Rod, Wire and Special Shapes, Rolled, Drawn, or Cold Finished
QQ-P-416	-	Plating, Cadmium (Electrodeposited)
QQ-W-423	-	Wire, Steel, Corrosion-Resisting
QQ-Z-325	-	Zinc Coating, Electrodeposited, Requirements For
V-T-276	-	Thread, Cotton
WW-T-700/6	-	Tube, Aluminum Alloy, Drawn, Seamless, 6061

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MILITARY

MIL-F-495	-	Finish, Chemical, Black, For Copper Alloys
MIL-W-530	-	Webbing, Textile, Cotton, General Purpose, Natural Or In Colors
MIL-T-3530	-	Thread And Twine, Mildew-Resistant Or Water Repellant Treated
MIL-A-8625	-	Anodic Coatings, For Aluminum And Aluminum Alloys

STANDARDS

FEDERAL

FED-STD-595	-	Color
FED-STD-751	-	Stitches, Seams and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables For Inspection By Attributes
MIL-STD-129	-	Marking For Shipment and Storage

DRAWINGS

Defense Personnel Support Center (DPSC)

10144	-	SPLINT, LEG, THOMAS, HALF-RING, ALUMINUM

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D2240-68	-	Indentation Hardness Of Rubber And Plastics By Means Of A Durometer, Test For

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

COLOR ASSOCIATION OF THE UNITED STATES, INC.

United States Army Standard Color Card For The Official Standardized Shades Of Sewing Threads

(Application for copies should be addressed to the Color Association of the United States, 200 Madison Avenue, New York, NY 10016.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Half-ring, extension rod, shaft sleeves, locking collars, locking washer, and connecting lugs. The half-ring extension rod, shaft sleeves, locking collars, locking washers and connecting lugs shall be aluminum alloy 6061, temper T6 in accordance with QQ-A-225/8.

3.1.2 Long and short shafts. The long and short shafts shall be aluminum alloy 6061, temper T6 in accordance with WW-T-700/6.

3.1.3 Lock spring. The lock spring shall be corrosion resistant, spring temper steel wire, condition B in accordance with QQ-W-423.

3.1.4 Rivets. The rivets used to connect the connecting lugs to the half-ring shall be carbon steel. The rivets used to connect the connecting lugs to the long and short shafts shall be 2017-T4 aluminum alloy.

3.1.5 Screws and lockwashers (for locking washers). The screws and lockwashers used to fasten the locking washers to the extension rod shall be carbon steel.

3.1.6 Cushion. The cushion shall be homogeneous, noncellular, molded vinyl plastic.

3.1.7 Webbing. The webbing shall be type II, medium weight (hard texture), class 4 webbing in accordance with MIL-W-530.

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3.1.8 Buckle. The buckle shall be brass.

3.1.9 Thread. Thread shall be in accordance with V-T-276. The bobbin thread shall be type IIIA, heavy thread, 3 ply, ticket No. 10 and the needle thread shall be type IIIB, heavy thread, 3 ply, ticket No. 10.

3.1.10 Tubing for attaching straps to half-ring. The tubing shall be cold-drawn annealed steel tubing.

3.2 Design and construction. The splint shall be designed and constructed in accordance with DPSC Drawing 10144 and as specified herein.

3.2.1 Assembly of components. Assembled components shall be finished smooth and shall be free of cracks or other irregularities. Screws and rivets shall be set securely. The finished half-ring shall be capable of rotating freely in each direction to approximately 90°

3.2.2 Cushion. The cushion shall be one-piece seamless molded construction. The openings where the straps and splint rod emerge (four openings), shall be continuous and smooth. The fit between the cushion and the half-ring shall be tight.

3.2.3 Straps. There shall be two straps on each splint. The webbing used for the straps shall be 1-1/2 inches wide, uniformly woven and evenly finished with selvaged edges. Each strap shall be securely attached to the half-ring by means of a round, seamless tube. The tube shall be crimped directly over each groove in the half-ring and shall have chamfered, or rounded, smooth, inner edges to avoid possible tearing of the strap. A three prong buckle with guard shall be securely sewn, as shown on the applicable drawing, to the strap on the longer side of the splint. Stitching shall be as specified in 3.2.4. The free end of the strap on the shorter side of the splint shall have a diagonal cut as specified on the applicable drawing.

3.2.4 Stitching on webbing. All stitching on webbing shall be stitch type 301 in accordance with FED-STD-751. There shall be not less than 10 stitches per inch. All stitching shall be backstitched at the ends to prevent raveling.

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3.3 Performance.

3.3.1 Drop test. There shall be no signs of damage to the plastic cushion as a result of the drop test of 4.4.1.

3.3.2 Hardness. The plastic cushion shall have a durometer hardness of 21 ± 5 when tested as specified in 4.4.2.

3.3.3 Deterioration. After being conditioned at high temperature and allowed to cool to standard atmospheric conditions as specified in 4.4.3, the plastic cushion shall show no signs of tackiness, blocking or deterioration.

3.3.4 Flammability. The plastic cushion shall be self-extinguishing within five seconds when tested as specified in 4.4.4.

3.3.5 Recovery time. The plastic cushion shall show no evidence of marked or permanent indentation when tested as specified in 4.4.5. Cushions showing slight indentation 24 hours after completion of the test shall be considered acceptable.

3.3.6 Security of strap attachment. Both straps shall withstand a 150 pound pull without becoming detached from the splint, when tested as specified in 4.4.6.

3.4 Finish and color.

3.4.1 Cushion. Any flash on the plastic cushion shall be wet buffed to give the cushion a smooth surface. The color of the finished cushion shall be black or olive drab approximating color 34087 of FED-STD-595 (see 6.3).

3.4.2 Webbing. Webbing for the straps shall be vat-dyed olive drab No. 7 and treated with mildew inhibitor in accordance with class 4 webbing of MIL-W-530, except that the copper 8-quinolinolate shall be a minimum of 0.13 percent to a maximum of 0.23 percent. The diagonally cut free ends of the straps shall be dipped in a vinyl resin for the entire length of the cut.

3.4.3 Buckle. The buckle shall have a black chemical (oxide) finish in accordance with MIL-F-495.

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3.4.4 Thread. Thread shall be vat-dyed olive drab shade S-1 conforming to U.S. Army Standard Color Card. Thread shall be treated with mildew inhibitor in accordance with type I, class I thread of MIL-T-3530.

3.4.5 Aluminum parts. All aluminum parts shall be anodized in accordance with type II, class 2 of MIL-A-8625. The finish shall be olive drab color approximating color 34087 of FED-STD-595 (see 6.3). The shaft sleeves, and long and short sleeves shall be anodized after brazing has been completed. The connecting lugs shall be riveted to the long and short shafts after anodizing.

3.5 Health hazard. The plastic cushion shall be nontoxic and shall not constitute a health hazard to personnel. When tested as specified in 4.5.2, there shall be no evidence of inflammation or other adverse skin reactions caused by the material on the human subjects used in the test.

3.6 Identification marking. Splints shall be permanently and legibly marked with the manufacturer's name or registered trademark. The marking shall appear on a solid metal component of the splint.

3.7 Workmanship. Workmanship shall be first class throughout. The splints shall be free from burs, sharp edges, cracks or other defects which detract from their appearance or may impair their serviceability. All brazing shall be continuously even, smooth and free from flaws and brazing marks.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality (materials). Certificates of quality supplied by the manufacturers of the metals, vinyl, webbing and thread may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.1.4 Certificate of compliance (skin sensitivity patch test). Unless otherwise specified, the contractor shall certify for each lot of splints that the plastic cushions used in the fabrication of the splints are nontixic and meet the requirements of 3.5.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in table I. The unit of product for examination purposes shall be one splint.

TABLE I. Sampling for examination.

Type defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

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4.2.2 For tests on finished product. Sampling for tests on the finished product shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The unit of product for test purposes shall be one splint. A minimum of two samples shall be tested regardless of lot size. Acceptance number shall be zero for all sample sizes. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

TABLE II. Sampling for tests on finished product.

Characteristic	Requirement	Test procedure	Inspection level
Drop test	3.3.1	4.4.1	S-1
Hardness	3.3.2	4.4.2	S-2
Deterioration	3.3.3	4.4.3	S-2
Flammability	3.3.4	4.4.4	S-1
Indentation	3.3.5	4.4.5	S-2
Security of strap attachment	3.3.6	4.4.6	S-2

4.2.3 For tests on components. Sufficient samples of the components shall be selected from each lot of material used in the manufacture of the splints to perform tests as required by the applicable subsidiary specifications or by tests specified herein. When sample size for components covered by the subsidiary specifications is not specified, the sample size for each test shall be as indicated for inspection level S-1 in accordance with MIL-STD-150, except that the acceptance number shall be zero for all sample sizes, and a minimum of two samples shall be taken regardless of lot size.

4.3 Examination.

4.3.1 Splints. The splints shall be examined to determine compliance with the requirements contained in this specification. The splint shall be examined for possible defects, including, but not restricted to, those listed in table III.

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TABLE III. Classification of defects.

ıtion	Defect
	Parts missing or not correct size. Parts not made of specified material. Parts not secured as required. Cushion not seamless, molded one piece construction. Tubing bent, dented, cracked or otherwise damaged. Half-ring cannot be rotated easily. Stitching on webbing not as specified.
	Finish of any part not as specified. Strap on short side of splint does not have diagonal cut or vinyl dip. Buckle not design specified. Parts not free of flash, burs, or sharp edges.

ımination of preparation for delivery. An examination ıade to determine whether the packaging, packing, and ɔmply with the requirements of section 5. Defects ɜcored as specified in table IV. Sampling shall be in ɜ with MIL-STD-105. The sample unit shall be one fully prepared for delivery. The lot shall be the containers offered for inspection at one time. The ı level shall be S-2 and the acceptable quality .1 be 4.0 defects per hundred units.

ɜ IV. Examination of preparation for delivery.

ɔn	Defect
ɜ	Not as specified.
	Quantity in interior packages and shipping containers not as specified.
and packing	Any component missing or damaged.
.p	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
	Omitted, incorrect, illegible, improper location, sequence, or method of application.

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4.4 Tests on the finished product. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Drop test. After conditioning for four hours at minus 65° Fahrenheit (F), the splint shall be held vertically with the half-ring downward, and shall be dropped from a height of three feet onto a concrete block, measured from the lowest point on the half-ring. The same test shall be repeated with the half-ring horizontal and at 45° to each side, in turn. The plastic cushion shall meet the requirements of 3.3.1.

4.4.2 Hardness. Hardness of the cushion shall be tested in accordance with ASTM Standard D2240-68, except that the test sample paragraph does not apply. The indentation point of the durometer shall be made at the highest point of the curve of the sample and the hardness reading taken when the entire length of the presser foot just makes contact with the sample. The average of 10 readings shall be used to determine compliance with 3.3.2.

4.4.3 Deterioration. The test sample shall be conditioned in a circulating oven at 165° + 5° F for a minimum of 18 hours and then allowed to cool to 70° ± 2° F at a relative humidity of 65 ± 2 percent. It shall remain at this latter condition for four hours. At the end of this time the cushion shall meet the requirements of 3.3.3.

4.4.4 Flammability. The test sample shall be suspended in a vertical plane, in a draft free area, with the ends of the plastic cushion pointing downward and equidistant from a horizontal plane. A Bunsen burner with a flame 1/2 to 3/4 inch in height shall be placed under one end of the cushion. The position of the Bunsen burner shall be adjusted so that the flame tip just makes contact with the plastic. At the end of 15 seconds, the flame shall be removed. By means of a stopwatch, the time for the plastic to stop burning shall be measured in order to determine compliance with 3.3.4.

4.4.5 Recovery time. Two 3/8 inch diameter steel rods shall be placed approximately 3-1/2 inches apart across the plastic cushion on the assembled splint. A five-pound weight shall be placed across these two steel rods. The splint, with test rods and weight, shall remain in position for 72 hours. The cushion shall be examined for compliance with 3.3.5.

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4.4.6 Security of strap attachment. With the splint secured, each strap shall be subjected to a 150 pound static load for a period of 1/2 hour, tangent to the half-ring at the point of attachment to the half-ring to determine compliance with 3.3.6.

4.5 Tests on components.

4.5.1 Webbing vinyl-dip. The vinyl on the diagonal cut of the free end of the longer strap shall not become tacky after being subjected to 165° ± 5° F for four hours; shall not crack or peel when the vinyl dipped end is flexed back and forth three times after exposure to minus 65° ± 5° F for one hour and while still at this temperature; and shall not swell up or become gelatinous after four hours exposure to an atmosphere of 100 percent relative humidity at 100° F.

4.5.2 Skin sensitivity patch test. One-half (1/2) inch square of the finished plastic cushion of the splint shall be applied to the skin of at least 10 human subjects (not less than one square applied to each subject in such a manner that the outside surface of the cushion material is applied to the (skin). This application shall be made by means of adhesive plaster and shall remain in place for 72 hours except for removal to permit examination of the skin test area after every 24 hours (within that 72-hour period) for any evidence of inflammation or other adverse skin reactions. If the material unduly irritates the skin, the test patch shall be removed immediately.

5. PACKAGING

5.1 Preservation packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Each splint shall be wrapped in kraft paper or other noncorrosive material. The wrap shall be secured with kraft or other suitable tape.

5.1.1.2 Intermediate package. Two unit packages shall be packaged in a snug fitting box constructed in accordance with PPP-B-636, class domestic. Closure of the box shall be adequate to prevent accidental opening.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used for the purpose which will afford the degree of protection required for shipment and use of the end product for its intended purpose within the continental limits of the United States of America.

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5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Twelve intermediate packages (24 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. Closure, waterproofing and strapping shall be as specified in the appendix of PPP-B-636.

5.2.2 Level B. Twelve intermediate packages (24 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class domestic. Closure shall conform to method I or II as specified in the appendix of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The splint covered by this specification is suitable for use in immobilization of fractures and treatment of injuries to the lower extremities of personnel.

6.2 Ordering data. Procurement documents should specify:

a. Title, number and date of this specification.
b. National stock number.
c. The applicable level of packaging and packing (see 5.1 and 5.2).

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6.3 Availability of color chips. Individual chips of the colors specified in 3.2.4 are available from:

Commissioner
Federal Supply Service
General Services Administration
Standardization Division, Paints Branch
Washington, DC 20406

6.4 This specification is intended to cover the following item appearing in the Federal Supply Catalog:

National Stock No.	Item Identification
6515-00-372-5100	SPLINT, LEG, THOMAS, HALF-RING, ALUMINUM

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project No. 6515-1958

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE
MIL-S-36040A SPLINT, LEG, THOMAS, HALF-RING

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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